

Depth-Damage Curve Model (MKM-DD-001)

Wolfram Oracle Benchmark of the Flood Risk Model Document

MKM Research Labs

Version 0.1 — 03-November-2026

David K Kelly

CONFIDENTIAL — PROPRIETARY

Legal Notice

PROPRIETARY AND CONFIDENTIAL

This document, including all algorithms, models, methodology, formulas, calculations, and intellectual concepts contained herein, constitutes the exclusive intellectual property and confidential trade secrets of MKM Research Labs ("MKM").

Any usage, reproduction, distribution, modification, or disclosure of this document or any portion thereof, without the express prior written authorisation from MKM Research Labs is strictly prohibited and constitutes an infringement of intellectual property rights.

The document is provided on an "as is" basis. MKM Research Labs makes no representations or warranties of any kind, express or implied, regarding its accuracy, reliability, or suitability for any particular purpose.

All rights reserved. © 2021–2026 MKM Research Labs.

Governed by the laws of the United Kingdom.

Contents

1 Executive Summary **3**

2 Model Purpose and Scope **4**

2.1 What is benchmarked 4

2.2 What is not 4

3 Mathematical Framework **4**

4 Benchmark Method **4**

5 Validation and Backtesting **5**

5.1 The model document's claims 6

6 Model Limitations and Known Weaknesses **6**

7 Document History **7**

1 Executive Summary

This document benchmarks the Depth-Damage Curve Model (MKM-DD-001), whose filed model document is *Flood Risk Model* (`flood_risk.pdf`, v1.1, 18-Feb-2026). It turns flood depth into a damage ratio, and carries the velocity, attenuation and hydrograph functions that the property flood propagation and the hazard curve builder use. It asks whether the code computes what the filed document says, and whether what the document says holds.

The code computes the formulae it still implements. The depth-damage curve, Manning's velocity, slope, travel time, attenuation, haversine, IDW and correlation matrix agree with a 50-digit implementation of the document's equations to 3.2×10^{-16}. The hydrograph agrees to the code's printed precision. The portfolio Monte Carlo's mean loss is within 1.82 standard errors of the exact expected loss (Table 1). Every printed figure in Tables 5–7 is reproduced. Wolfram, computing each independently, agrees with the reference to 4.1×10^{-31}.

But the filed document no longer describes the model. Its source has since reached v1.4, which was never filed, and the code has moved past both (Table 2):

1. **The attenuation length is 10,000 m, not 2,000 m.** Table 2 and §3.3.4 give $\ell = 2{,}000$ m; the code's default is 10,000 m, and its own docstring says 3,000. At 2 km a property keeps 82% of the flood signal, not 37%.

2. **Equations 2 and 4–6 are not implemented.** The property-type factor and the depth-from-the-Thames-point calculation were removed from the code. Their constants remain in config, unread.

3. **Eq. 15's attenuation is not applied in the hydrograph.** The caller applies it to the peak instead. Applied as written, it would lower peak depth by up to 1.2 m.

And some of its figures are wrong.

1. **Table 9's velocities are a third of Manning's.** At depth 1 m and slope 0.003, Eq. 11 gives 2.19/1.37/0.91/0.55 m/s against the printed 0.64/0.40/0.27/0.16.

2. **The ±8% headline is not a property of the model** (§1, Table 8, Q9). A 10% shift of the curve moves expected loss by exactly 10% wherever the clip at 1 does not bind. On the benchmark portfolio it is 1.064×, so the figure depends on the portfolio.

3. **One degree is not "about 2%" out east–west at 51.5°N** (§8): it is 61%.

Evidence. 14 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `7bbb63078a59`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-11-03 by `python -m tools.wolfram_oracle dd001`.

2 Model Purpose and Scope

2.1 What is benchmarked

The four modules of `src/models/floodrisk/` the document describes, imported as production imports them:

- `depth_damage.scalar_depth_damage`, which the hazard curve builder and the property flood propagation call;
- `velocity`: Manning, slope, travel time, attenuation and the property hydrograph, which the propagation and the terrain grid call;
- `spatial`: haversine, IDW and the correlation matrix;
- `risk_analytics`: the Monte Carlo, VaR, ES and HHI. These, IDW and the correlation matrix have no caller in `src`.

2.2 What is not

The calibration of the curve to survey data (§4.2, §7), the backtests of §7, and Tables 11–12, which need a portfolio the document does not specify. Also out of scope: KMeans clustering, and the degree-5 curve in config, except for the fit its own comment states. This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

The reference implements the filed document's equations directly from its text:

- Table 1's curve, interpolated linearly (Eq. 1);
- Manning's $v = h^{2/3}S^{1/2}/n$ with $S \geq 0.001$ (Eqs. 11–12), and $t = d/(3600v)$ (Eq. 13);
- the attenuation $e^{-d/\ell}$ (Eq. 14);
- the hydrograph of Eqs. 15–17;
- the haversine (Eq. 7) and IDW (Eq. 8);
- the correlation $\rho_0 e^{-111 d_{\text{deg}}/(L/1000)}$ with a unit diagonal (Eq. 10).

The expected loss, exactly. Eq. 18 shocks each damage ratio to $\text{clip}(\delta(1 + 0.2z), 0, 1)$ with z standard normal. Since $X = \delta(1 + 0.2z) \sim N(\delta, (0.2\delta)^2)$,

$$\mathbb{E}\,\text{clip}(X, 0, 1) = \delta\big[\Phi(\beta) - \Phi(\alpha)\big] + 0.2\delta\big[\varphi(\alpha) - \varphi(\beta)\big] + 1 - \Phi(\beta), \quad \alpha = -5,\ \beta = \tfrac{1-\delta}{0.2\delta}. \tag{1}$$

The expected portfolio loss is $\sum_j V_j\, \mathbb{E}\,\text{clip}(X_j, 0, 1)$, whatever the correlation. So a 10% shift of the curve scales it by exactly 1.10 until β is small enough for the clip to bind, around $\delta > 0.6$.

4 Benchmark Method

The cases.

- Every family is evaluated over a grid covering the document's tables and their edges: depths from -0.2 to 7 m, zero and negative roughness, slopes below the floor, and negative distances.
- The hydrograph is run on a 168-hour triangular storm at three travel times.

- The Monte Carlo runs on a 40-property portfolio near the Thames reference point, with depths spanning every damage band. It uses the production defaults: 1,000 draws, seed 42, and the code's own correlation matrix.

The printed tables. Tables 5–7 are judged as printed. The deleted generator rounded each value and then formatted it by size, and the code's value, treated the same way, must give each printed figure.

Wolfram. Wolfram works from each input's exact value, independently of the reference:

- the curve, through its own first-order `Interpolation` of Table 1;
- Manning and the attenuation;
- the great-circle distance, as R times the angle between the two points' unit vectors;
- the clipped expectation, by `Expectation`;
- the least-squares fit of a degree-5 curve through the origin to Table 1;
- the true length of 0.1° east–west at 51.5°N, by `GeoDistance` on WGS84.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Eq. 1, the depth-damage curve: code against the reference, worst relative error	2.2×10^{-16}	1.0×10^{-12}	pass
Eq. 11, Manning's velocity: code against the reference, worst relative error	1.9×10^{-16}	1.0×10^{-12}	pass
Eq. 12, the ground slope: code against the reference, worst relative error	5.6×10^{-17}	1.0×10^{-12}	pass
Eq. 13, the travel time: code against the reference, worst relative error	2.9×10^{-16}	1.0×10^{-12}	pass
Eq. 14, the attenuation: code against the reference, worst relative error	3.2×10^{-16}	1.0×10^{-12}	pass
Eq. 8, IDW: code against the reference, worst relative error	8.2×10^{-17}	1.0×10^{-12}	pass
Eq. 13: no velocity gives an infinite travel time	yes		pass
Eq. 7, the haversine: code against the reference, worst error (m)	9.0×10^{-10}	1.0×10^{-6}	pass
Eq. 10, the correlation matrix: worst relative error	3.1×10^{-16}	1.0×10^{-12}	pass
Eqs. 15–17, the hydrograph without a: worst error (m)	7.5×10^{-5}	1.0×10^{-4}	pass
Eq. 18–19: Monte Carlo mean loss (1,000 draws) against the exact, $\lvert z \rvert$	1.82	4	pass
Eq. 21: the code's ES_{99} against its own draws	0	1.0×10^{-12}	pass
Wolfram agrees with the reference	4.1×10^{-31}	1.0×10^{-25}	pass
The code's degree-5 curve is the exact least-squares fit to Table 1	4.4×10^{-9}	5.0×10^{-9}	pass

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
Table 5	depth-damage curve, 13 depths	13 of 13 figures reproduced	met
Table 6	Manning's velocity	18 of 18	met
Table 7	attenuation	24 of 24	met
§3.3.4, Table 10	$e^{-1} \approx 0.37$, $e^{-3} \approx 0.05$; 0.14, 0.37, 0.67	as printed	met
Table 9	velocity 0.64/0.40/0.27/0.16 m/s at $d = 1$, $S = 0.003$	Eq. 11 gives 2.19/1.37/0.91/0.55; the travel times imply 2,995–3,024 m at the printed speeds	**not met**
Table 8, §1, Q9	a ±10% curve shift moves expected loss 1.08×/0.92×, tail risk ±8%	exactly 1.10× wherever the clip does not bind (to 1.100 at $\delta = 0.6$, 1.056 at 0.9); 1.064/0.916 and VaR_{99} +4.8% on the benchmark portfolio: a property of the portfolio, not the model	**not met**
§8(8)	1° ≈111 km is about 2% out east–west at 51.5°N	60.6% ($1/\cos 51.5° - 1$); on WGS84, 0.1° is 6,944 m, not 11,100	**not met**
§3.3.4, Table 2	attenuation length ℓ = 2,000 m	the code's default is 10,000 m, its docstring says 3,000	**not met**
Eqs. 2, 4–6	property-type factor; depth from the Thames point with linear decay	not in the code: `PROPERTY_TYPE_DAMAGE_FACTORS`, `FLOOD_MAX_DISTANCE_M` are never read	finding
Eq. 15	the hydrograph scales by the attenuation a	the code does not (the caller does); applied as written, peak depth falls by up to 1.2 m	finding
§3.2.4, §5.4	the correlation matrix is positive definite	smallest eigenvalue 0.567	met
§7.4, §9.3	correlated shocks by Cholesky	NumPy's `multivariate_normal` factorises by SVD	finding
§3.4.3	HHI = $1/N$ for an equal portfolio	0.0250 for $N = 40$	met
§9.3	about 800 MB for 10,000 properties	800 MB	met
config	the undocumented degree-5 curve: R^2 0.999244, RMSE 0.009368, max error 0.020388, monotone	0.999244, 0.009368, 0.020388; least slope 0.052	met

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute what it implements of the document; whether the curve fits UK damage is §7's question.

2. **The document.** The filed v1.1 should be replaced. Its source reached v1.4 (26-Mar-2026, ℓ = 10,000 m) and was never filed, and even v1.4 predates the removal of Eqs. 2 and 4–6, the single-gauge propagation of the code's v2.1, and the degree-5 curve. A reissue should document what the code does, correct Table 9, §8(8) and the ±8% headline, and say that Monte Carlo draws use SVD (§7.4, §9.3).

3. **The attenuation length.** `compute_retention`'s docstring should agree with its 10,000 m default.

7 Document History

Date	Version	Description	Author
03-Nov-2026	0.1	First issue	David K Kelly